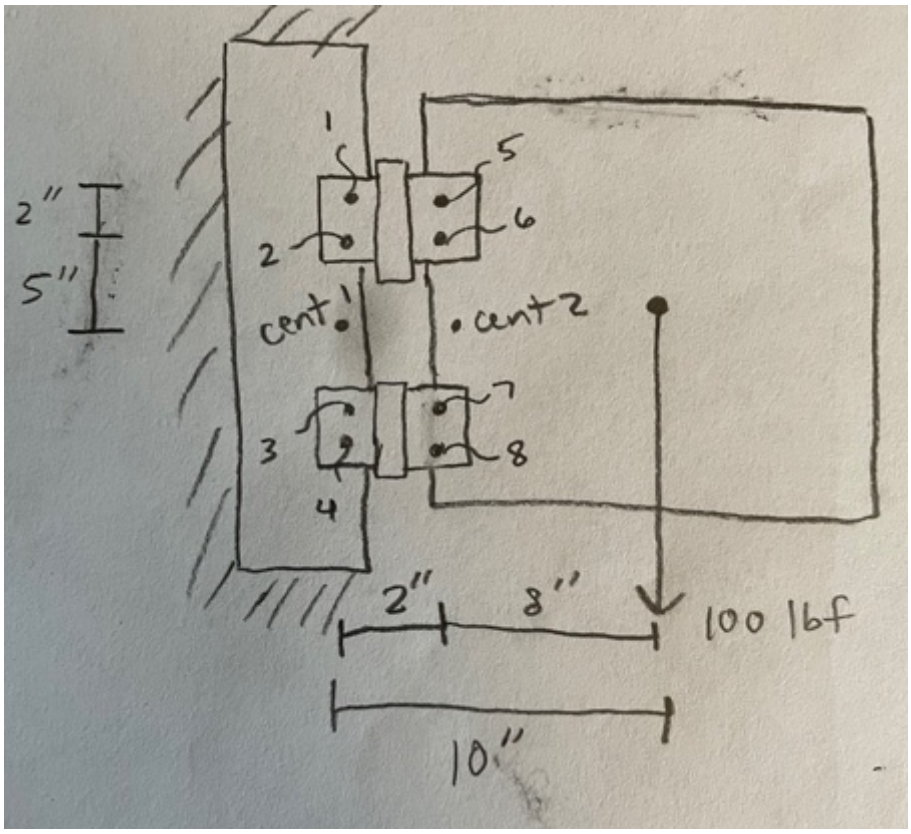
Engineering

How do you calculate the shear force on the screws that connect a door to a jamb via hinges

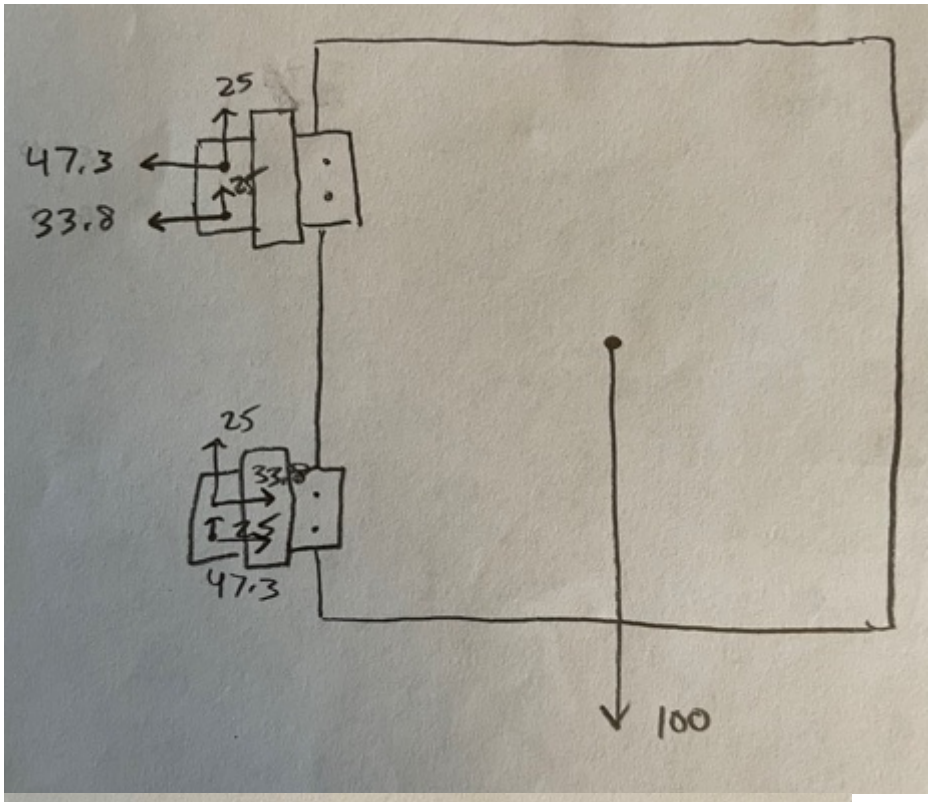
Asked 1 year ago Modified 1 year ago Viewed 578 times

2

I'm trying to calculate the shear forces on hinge screws or bolts. Here is a schematic of the door with the relevant values:



First, I took a free body diagram of the door and its hinges. The calculations to find the forces are also shown below. The forces consist of direct shear which is just the total weight divided by the number of bolts (4) and shear from torsion about the centroid of the bolt group.

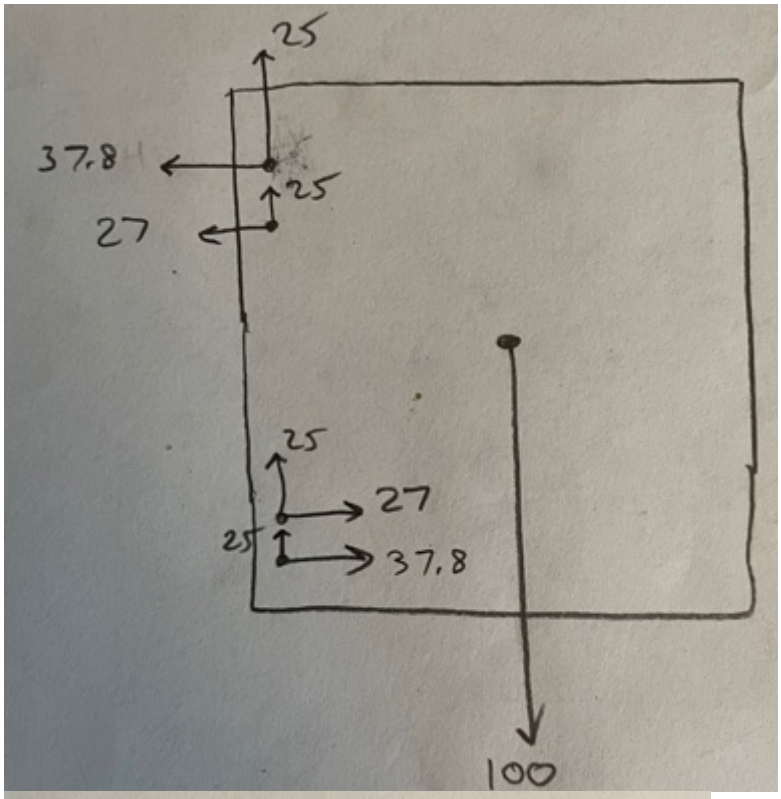


$$F_s = \frac{100\,\text{lbf}}{4} = 25\,\text{lbf}$$

$$F_{1,T} = F_{4,T} = \frac{100\,\text{lbf} \cdot 10.5 \cdot 7.5}{2(5^2+7^2)} = 47.3\,\text{lbf}$$

$$F_{2,T} = F_{3,T} = \frac{100 \cdot 10.5}{2(5^2+7^2)} = 33.8\,\text{lbf}$$

Next, I took the free body diagram of just the door and calculated the forces a similar way:

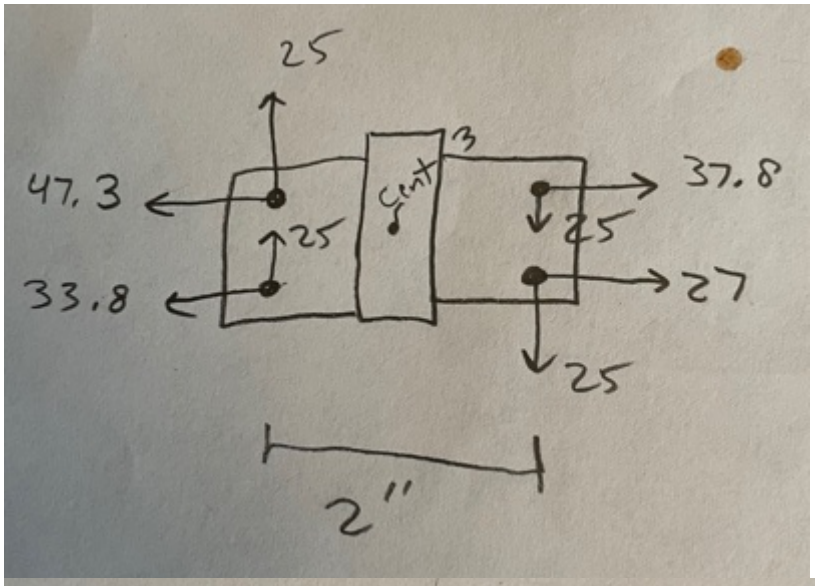


$$F_s = \frac{100}{4} = 25 \text{ lbf}$$

$$F_{5,T} = F_{8,T} = \frac{100 \cdot 8.7}{2(5^2+7^2)} = 37.8 \text{ lbf}$$

$$F_{6,T} = F_{7,T} = \frac{100 \cdot 8.5}{2(5^2+7^2)} = 27.0 \text{ lbf}$$

Where I get confused is when I take the free body diagram of just a single hinge and sum the moments about its center of mass, I get a non-zero moment, which doesn't make sense to me.



$$\Sigma M_{cent} = 2[(25+25)\cdot 1] + 37.8\cdot 1 + 33.8\cdot 1 - 47.3\cdot 1 - 27\cdot 1$$

$$= 97.3 \text{ lbf-in}$$

Can someone explain to me where my approach or thinking is incorrect?

statics torque moments bolting shear

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asked May 23, 2021 at 20:30



Ryan
33 3

This is a nice problem, I'll look into it if you don't get another response (it will take me some time to look into it), however, I thing that I noticed about the centroid of a joint is that the forces on the x axis do not add up to zero. Which means that the joint (and thereofre the door should be moving to the left). (If I remember correctly - I had done things like that a few years ago - obtaining the forces is a non intuitive way) – NMech May 24, 2021 at 12:08

@NMech I'm still a bit confused about why my forces aren't balancing. There must be something about this problem that distinguishes it from fastener group problems, but it's not clear to me what this is. – Ryan May 25, 2021 at 0:12

2 Answers

Sorted by:

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2

The shear stress on hinge screws is close to zero. The door is held by the friction of the plate to the frame, not by the shear strength of the screws. All your calculations ignore that the screws are holding the hinge plate fast to the frame. If you calculated the force holding the plate to the frame you would find that the friction is a very large force. This is without considering that hinges are generally sunk into the frame and are supported by the edge of the plates as well.

This is the same concept used in automobile wheel lug nuts - the nuts are not in shear because the load is held by the friction between hub and wheel. Lug nuts in shear would be almost certain to fail due to fatigue.

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answered May 26, 2021 at 2:03



Tiger Guy

4,628 4 15

I agree that my example is not realistic. However, I'd still like to know why the forces don't balance in the frictionless case when using a fastener group approach. – Ryan May 26, 2021 at 10:45

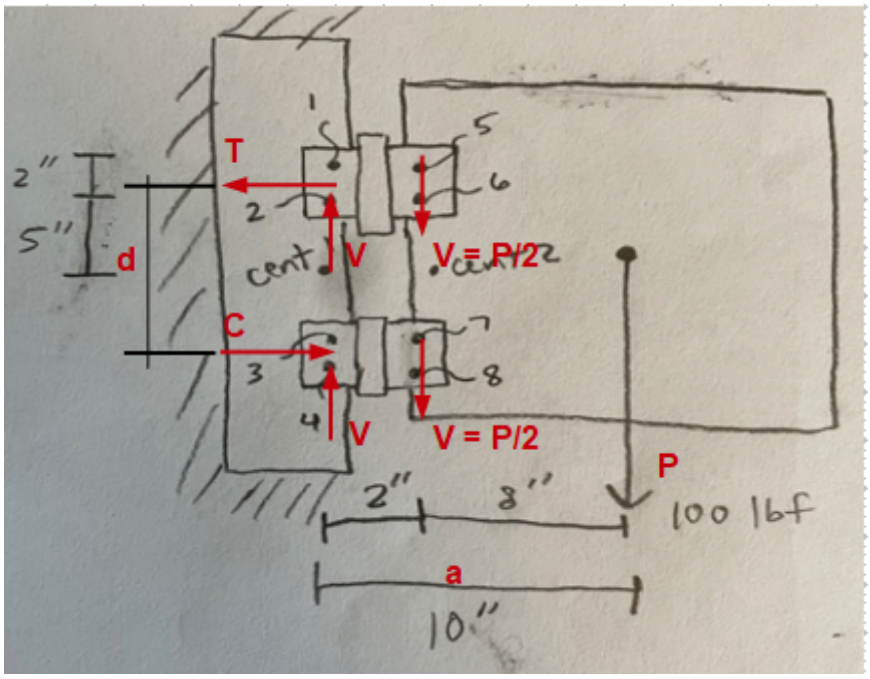
if the forces don't balance you've done it wrong. If the issue is there's still a moment, of course there's a moment, the forces are not in line. – Tiger Guy May 27, 2021 at 19:31

1

Method 1 - Common Structural Design Approach

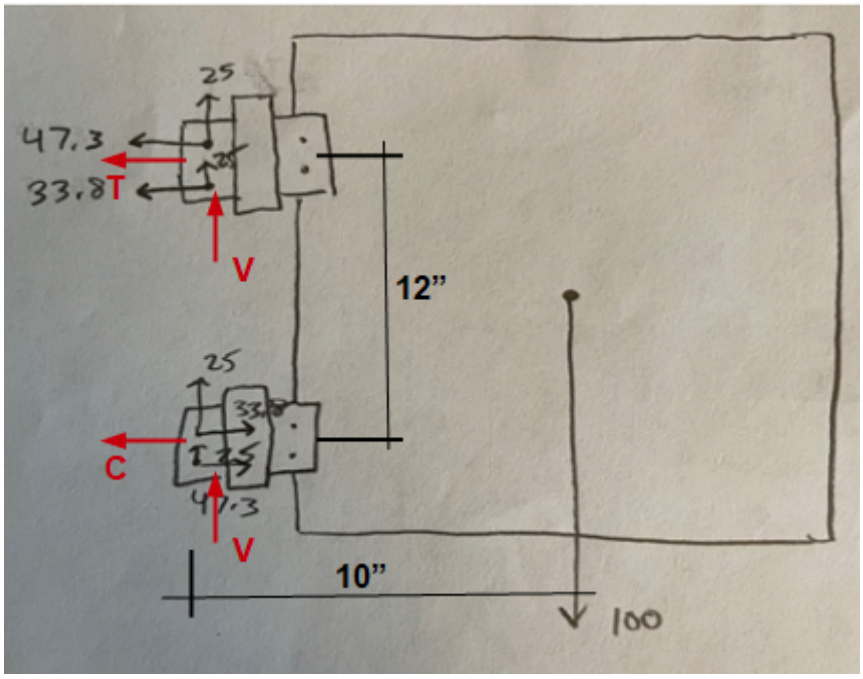
Assume the door is simply supported by the hinges on the jamb, and solving the reactions on the support hinges. The design shear on the screw is the resultant of T & V, which numerically equals the resultant of C & V.

For the purpose of design, you can stop here, since all screws must be the same size for the reason of economy, and with the understanding that the jamb side hinges are more critical than the door side hinges since they carry the loads passed from the centroid of the door to the door side hinges, then onto the jamb side hinges.



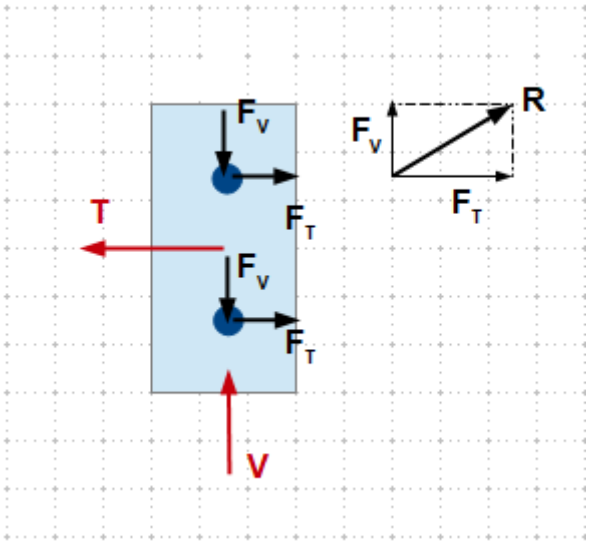
Let's see how this scheme works:

In the figure below, $\sum M = 0$, then $T = C = 100 * 10/12 = 83.33lbs$; and $\sum Fy = 0$, and assume the two hinges share the same amount of the vertical load, that is $V = 100/2 = 50lbs$.



Now the forces on the upper hinge are as shown in the figure below. The component forces on a single screw are simply equal to one-half of the respective force in the vertical and

horizontal directions ($F_V = V/2$ & $F_T = T/2$). Finally, the shear force on a single screw

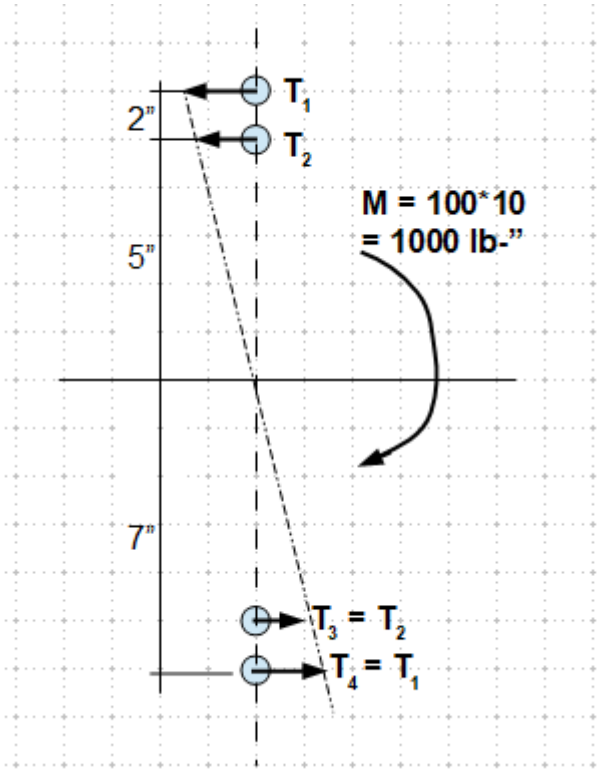


$R = \sqrt{F_V^2 + F_T^2}$

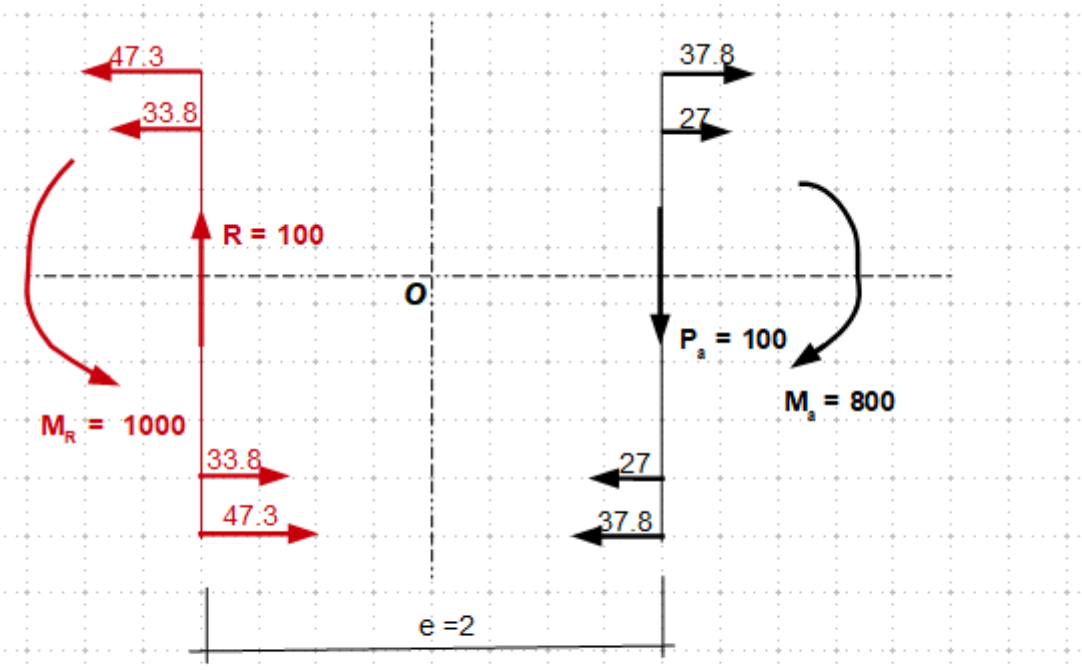
Method 2 - Exact Solution

As shown in the figure below, you can solve the component horizontal forces using the exact method, which assumes a rigid bolt group subjects to externally applied moment/torque, and the stress of the screws are linear elastic.

Through the realization of $T_4 = T_1$, $T_3 = T_2$ and T_2 is proportional to T_1, you can solve the component force by $\sum T_i * a_i = M$. The most critical shear force thus is the combined horizontal and vertical shears of the outermost screws.

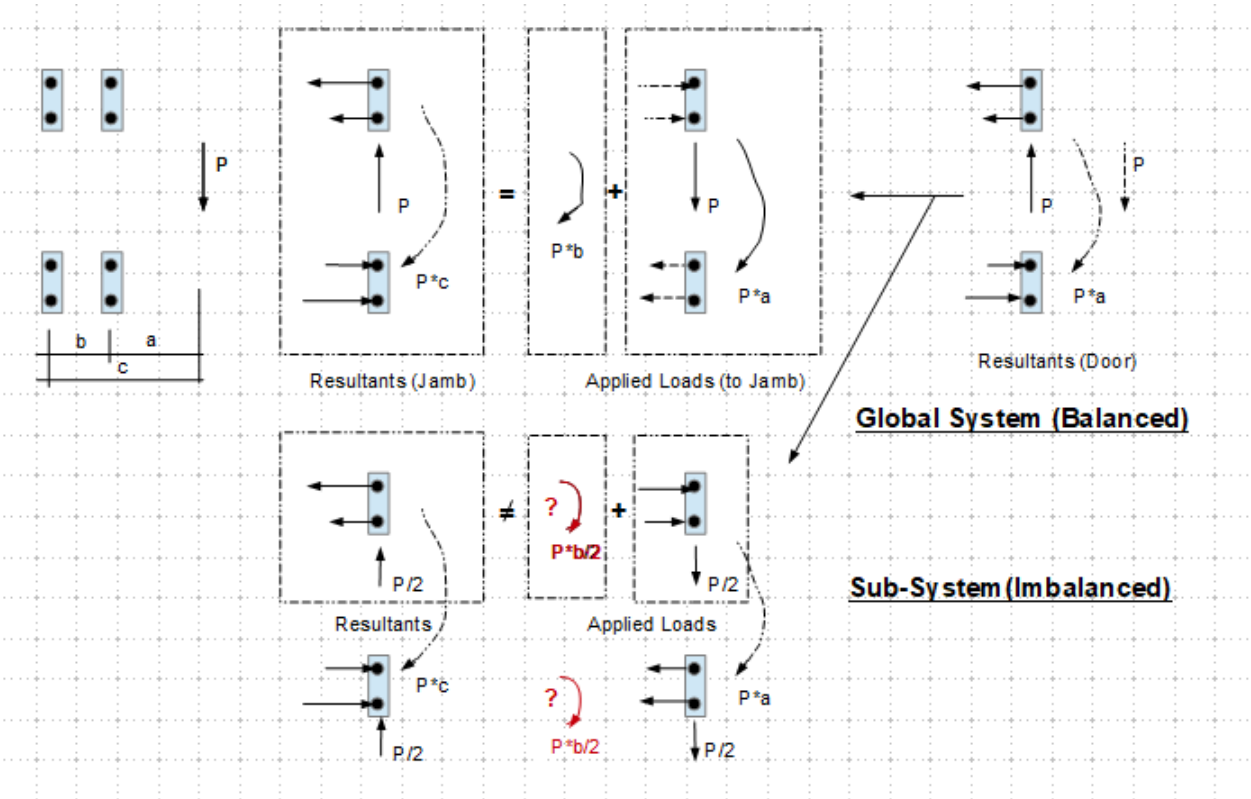


System Equilibrium Check - The system is in equilibrium, since $\sum F_x$, $\sum F_y$ and $\sum M$ about the center of origin "***O***" are all zero.



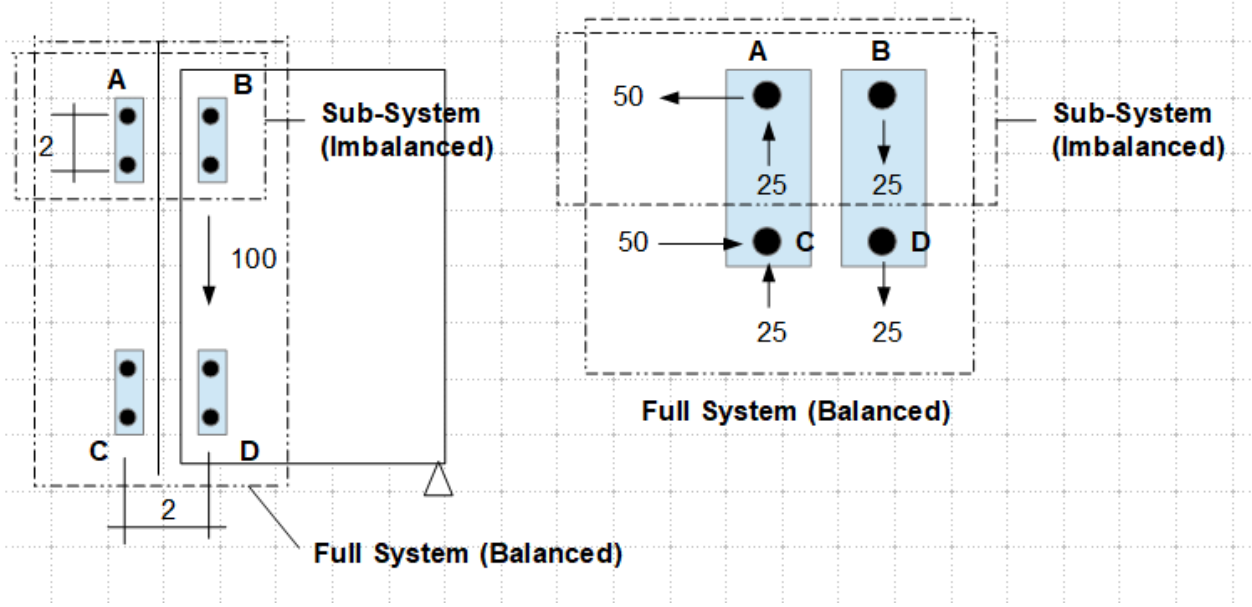
Note that you can't expect the unbalanced sub-system (the upper or lower pair) alone to be in equilibrium, because in which $\sum F_x \neq 0$ *and* $\sum M_{ctr} \neq 0$.

The sketch below demonstrates my last comment. Hope this helps.



Final THought

To further illustrate the concept of system balance, let's assume the door was supported on its right bottom corner, and the hinges on its left edge acting as another support, which subjected to a force of 100 lbs, the system is as the left figure shown in the sketch below.



Now we can analyze the upper pair of hinges separately, the respective forces are shown in the diagram on the right, and we note that while the entire system (which includes all four screws) is in the state of equilibrium, the sub-system does not (since $\sum F_x \neq 0$).

In a similar manner, let's imagine the hinge "A" is in place of the screw "A" and so on for hinges B thru D, then, through drawing a parallel with the note above, we realize why the pair of the upper hinges in your calculation ending in system imbalance.

Though I didn't check the number, I think your calculation and the results are theoretically correct, as there are indeed having unbalanced forces locally (sub-system), but will be balanced globally (full system).

Share Improve this answer Follow edited May 25, 2021 at 13:33 answered May 23, 2021 at 21:07



r13
7,466 3 6 25

Can you explain why my approach is invalid? I tried to approach it like a fastener group problem, but for some reason that doesn't seem to work here. – Ryan May 23, 2021 at 21:27

Note the hinge is essentially two leafs connected through a link, the system moment is P*a* + *P*b = P*L, in your example a = 8", b = 2", and L = 10". All horizontal forces on the upper hinges are pointing to one direction that in reverse to the forces on the lower hinge. The vertical forces on both hinges are all pointing up. From here, you can check the system equilibrium at any point with all forces included. Which should be zero of course. – r13 May 24, 2021 at 3:07

Hinges are set-in to the jamb and can hold doors with no screws in - due to the flat edges, so how does that change the analysis? – Solar Mike May 24, 2021 at 4:48

@Solar Mike There must be something attaching the hinge to the jamb, then the analysis may change slightly, but the concepts remain. – r13 May 24, 2021 at 14:24

@r13 thank you for your thorough answer. Your math makes perfect sense, but can you explain how a sub-system can have unbalanced forces? What is preventing the hinge from translating or rotating if its forces and moments are unbalanced? Is the issue that the problem is presented in an overly idealized

way (i.e., all bodies are rigid, friction between the hinges and jamb are ignored, etc.) to get a realistic answer? – Ryan May 25, 2021 at 0:04